Week 23: Sorting machines

Adult guide built from the problem files · Unpiloted. The table kit is physically unrehearsed.

Problems and routes

Problem	K–1	2–3	4–5	route
Which starts does the two-bar machine sort?	P1			first
Build a three-card sorter	P2			first
Starts that beat a machine	P3			first
Sort one set, fail the other	P4			first
One more bar	P5			return
Build a four-card sorter	P6			return
Which starting orders does each machine sort?		P1		first
The fewest bars for three lanes		P2		first
Repair a four-lane machine		P3		first
Two 0s and two 1s		P4		return
Sort every 0-1 start		P5		return
Neighboring bars only		P6		return
A failing start or a complete record			P1	first
A four-lane sorter			P2	first
Thresholds: numbers to 0s and 1s			P3	first
Does sorting 0s and 1s guarantee sorting numbers?			P4	first
Which machine sorts?			P5	first
Swap records			P6	return
The fewest bars			P7	return

First route. K–1 P1–P4; Grades 2–3 P1–P3; Grades 4–5 P1–P5. **Return visits.** K–1 P5–P6; Grades 2–3 P4–P6; Grades 4–5 P6–P7.

Key, one entry per problem

Which starts does the two-bar machine sort?

K–1 P1 · first route

Idea. A fixed machine can sort some starts and not others: bars (1,2) then (2,3) sort four of the six orders of three cards and fail 2,3,1 and 3,2,1.

Build a three-card sorter

K–1 P2 · first route

Idea. Three bars, such as (1,2), (2,3), (1,2), sort every order of three cards; no two-bar machine does.

Starts that beat a machine

K–1 P3 · first route

Idea. Bars (1,2), (2,3), (1,2) sort every start; bars (1,2), (1,2), (2,3) repeat a comparison and fail 2,3,1 and 3,2,1.

Sort one set, fail the other

K–1 P4 · first route

Idea. Bars (1,2), (2,3) sort every order of 0, 0, 1 but fail 1, 1, 0; bars (2,3), (1,2) sort every order of 0, 1, 1 but fail 1, 0, 0.

One more bar

K–1 P5 · return route

Idea. Each two-bar machine becomes a sorter with exactly the right last bar: (1,2), (2,3) and (1,3) respectively.

Build a four-card sorter

K–1 P6 · return route

Idea. Five bars, (1,2), (3,4), (1,3), (2,4), (2,3), sort every order of four cards; no four-bar machine does.

Which starting orders does each machine sort?

Grades 2–3 P1 · first route

Idea. Bars (1,2), (2,3) fail 2,3,1 and 3,2,1; bars (2,3), (1,2) fail 3,1,2 and 3,2,1; bars (1,3), (1,2), (2,3) sort all six orders.

The fewest bars for three lanes

Grades 2–3 P2 · first route

Idea. Three bars suffice for three lanes, and none of the nine two-bar machines sorts every order of 1, 2, 3.

Repair a four-lane machine

Grades 2–3 P3 · first route

Idea. The first two four-bar machines are repaired only by a last bar (2,3); the third, with (1,4) and (2,3), cannot be repaired by one bar.

Two 0s and two 1s

Grades 2–3 P4 · return route

Idea. Bars (1,2), (1,3), (2,3), (2,4) sort all six orders of 0, 0, 1, 1 but fail 0010, 0100, 1000 and 1110, so passing one kind of 0-1 start is not enough.

Sort every 0-1 start

Grades 2–3 P5 · return route

Idea. A four-lane machine can be checked on all 16 starts of 0s and 1s; the five-bar sorter passes them all.

Neighboring bars only

Grades 2–3 P6 · return route

Idea. With bars only between neighboring lanes, six bars are needed and enough: 4,3,2,1 has six out-of-order pairs and each neighboring swap fixes exactly one.

A failing start or a complete record

Grades 4–5 P1 · first route

Idea. Bars (1,2), (2,3) fail 2,3,1 and 3,2,1 (and 3,3,1 among the repeats); bars (1,3), (1,2), (2,3) sort every order, so the evidence must be one counterexample or all six starts.

A four-lane sorter

Grades 4–5 P2 · first route

Idea. Five bars, (1,2), (3,4), (1,3), (2,4), (2,3), sort every order of 1, 2, 3, 4.

Thresholds: numbers to 0s and 1s

Grades 4–5 P3 · first route

Idea. Replacing values by 0 (at most t) or 1 (larger than t) commutes with every bar, because min and max commute with an increasing map; a row with k different values gives k + 1 different 0-1 rows (4, 5 and 2 here).

Does sorting 0s and 1s guarantee sorting numbers?

Grades 4–5 P4 · first route

Idea. Zero-one principle: if a fixed machine left 9 above 4, the threshold t = 4 would give a 0-1 start that finishes 1 above 0, so a machine that sorts every 0-1 start sorts every start, repeats included.

Which machine sorts?

Grades 4–5 P5 · first route

Idea. Bars (1,2), (3,4), (1,3), (2,4), (2,3) sort every start; swapping the last two bars fails 0100 and 1000, so the 0-1 check settles both machines.

Swap records

Grades 4–5 P6 · return route

Idea. A swap/no-swap record fixes the rearrangement of distinct cards, so at most one start with that record finishes sorted; bars (1,2), (2,3) make four records and bars (1,2), (2,3), (1,2) make six.

The fewest bars

Grades 4–5 P7 · return route

Idea. Three lanes need 3 bars and four lanes need 5: a k-bar machine has at most 2^k records, each sorting at most one order, so 2^k must reach $3! = 6$ and $4! = 24$, and an exhaustive check rules out four bars.